Graphical Abstract

Machine Learning and Explainable Deep Learning for Forecasting Landslide Deformation using EGMS Time Series

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Slow-moving landslide deformation forecasting

Non-stationary, noisy and gappy cumulative InSAR displacement

Incremental deep learning framework

Causal preprocessing with detrending and slope encoding, followed by a CNN–BiLSTM–Attention model

Interpretable short-term forecasting

Stable 12-day-ahead deformation prediction with attribution highlighting recent states

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Machine Learning and Explainable Deep Learning for Forecasting Landslide Deformation using EGMS Time Series

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Abstract

Reliable forecasting of slow-moving landslide deformation remains challenging due to non-stationarity, noise, missing observations (gaps), and the cumulative nature of satellite-based displacement products. In this study, the time axis is defined on a fixed 12-day cadence, so the one-step forecast corresponds to a consistent 12-day horizon. Raw cumulative sequences are transformed into robust learning inputs through point-wise normalization using statistics computed exclusively from the training interval, local linear detrending, slope-encoded auxiliary channels, and outlier suppression. A hybrid model combining multi-scale convolutional filters, stacked Bidirectional LSTM layers, and a self-attention block is trained in an incremental next-step formulation, reducing drift accumulation while enabling short-term deformation forecasting. Evaluation on a strictly held-out temporal interval demonstrates stable generalization across heterogeneous kinematic regimes. Gradient-based attribution

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indicates that the network primarily relies on recent temporal states, yielding physically interpretable relevance patterns. Overall, the proposed framework provides a reproducible baseline for large-scale ground-motion forecasting and supports early evolution tracking of slow-moving landslide processes.
Keywords: Landslide forecasting; InSAR time series; Incremental deep learning; Attention mechanism; Explainable AI;

1. Introduction

Landslides are among the most pervasive geohazards worldwide, particularly in mountainous and tectonically active regions where slow but persistent ground deformation can progressively damage buildings, transportation corridors, and critical infrastructure [1]. The ability to monitor and forecast such long-term movements is essential for risk mitigation and sustainable land-use planning, especially in areas characterized by complex geological settings and widespread clay-rich formations [2]. Recent advances in satellite radar interferometry (InSAR) and the availability of continental-scale displacement datasets—most notably the European Ground Motion Service (EGMS)—have opened new opportunities for understanding slope instability over wide regions [3, 4]. However, effectively exploiting these dense but noisy time series for reliable forecasting remains a significant scientific and computational challenge, requiring both robust preprocessing and predictive models capable of capturing non-linear and non-stationary deformation behaviour [5, 6]. Slow-moving landslides are particularly problematic in the southern Apennines of Italy, where highly weathered clay formations and complex structural settings give rise to large, deep-seated landslide bodies [7]. Within the Basilicata

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region (Lucanian sector of the Southern Apennines), persistent ground deformation has been documented on both urbanized and natural slopes; the Latronico landslide is a representative example reported in the literature [8]. Such creeping behaviour, often modulated by seasonal hydrological variability and subsurface geological controls, motivates continuous monitoring and short-term forecasting for hazard management. As in many similar regions, traditional in-situ monitoring systems provide valuable information but lack the spatial density required to capture heterogeneous deformation patterns at regional scale. InSAR techniques have revolutionized the characterization of slow-moving landslides by enabling millimetric deformation measurements over large areas [9]. The EGMS, based on multi-temporal Sentinel-1 acquisitions, provides harmonized, quality-controlled displacement time series at unprecedented spatial density across Europe. These data have rapidly become a cornerstone for landslide detection, velocity estimation, and long-term deformation analysis. Nevertheless, EGMS time series remain challenging to model: they are often affected by atmospheric artefacts, temporal gaps, measurement noise, and strong non-stationary temporal dynamics [10]. Moreover, slow-moving landslides typically exhibit complex deformation regimes, including long-term creep, episodic accelerations, transient hydrologically driven slowdowns, and spatially variable kinematic behaviour—all of which complicate forecasting using classical statistical or physically based models.

In recent years, machine learning and deep learning have emerged as promising tools for modelling non-linear geophysical time series [11]. Recurrent neural networks such as Long Short-Term Memory (LSTM) and Bidirectional Long Short-Term Memory (BiLSTM) have demonstrated strong

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capabilities in capturing sequential dependencies [12], while convolutional neural networks (CNNs) can extract short-term deformation patterns and denoise local fluctuations [13, 14]. Attention mechanisms further enhance temporal modelling by allowing networks to identify and prioritize the most informative segments of the sequence [15, 16, 17, 18, 19]. Hybrid soft-computing frameworks that combine multiple deep-learning components have also been investigated to improve robustness and forecasting accuracy in complex time-series settings [20]. Despite these advances, existing works typically rely on local datasets, limited temporal records, or sparse ground-based monitoring systems. Only a few studies have explored the use of large-scale InSAR datasets for forecasting purposes, and even fewer have attempted to integrate hybrid deep-learning architectures with continental-scale products like EGMS [21]. Furthermore, the interpretability of deep learning predictions—critical for geoscientific adoption—remains a major open challenge, as model behaviour is often treated as a black box.

These issues highlight a clear research gap: few existing frameworks systematically integrate EGMS displacement time series with scalable deep learning architectures for deformation forecasting, and most prior studies focus on limited spatial extents, smaller datasets, or velocity estimation rather than short-term forecasting of incremental deformation [22, 23, 24]. Bridging this gap requires a dedicated modelling strategy that addresses noise, temporal gaps, non-stationarity, and spatial heterogeneity while delivering robust predictions suitable for real-world hazard mitigation. Compared with many InSAR-based landslide modelling studies, which focus on velocity estimation or operate on limited spatial extents [25, 26, 27], this study targets short-

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horizon forecasting from EGMS displacement time series at regional scale. We define the temporal index on a fixed 12-day cadence and formulate the learning problem as next-step increment prediction (delta mode) to mitigate drift accumulation in cumulative line-of-sight (LOS) products. To address the noise and gaps that characterize EGMS records, we adopt a strictly causal preprocessing and feature-construction scheme (train-only per-point standardization, within-window detrending, and a slope-encoded auxiliary channel, complemented by lightweight outlier robustification). Building on this representation, we evaluate a hybrid CNN–BiLSTM–Attention regressor on 594,978 EGMS measurement points using a temporally disjoint train/validation/test split, and complement accuracy metrics with attribution-based analyses (Integrated Gradients / Smooth Integrated Gradients with deletion–insertion tests) to characterize the temporal dependencies driving the forecasts.

In this work, we propose a comprehensive data-driven forecasting framework specifically designed for large-scale EGMS displacement time series and demonstrate its performance over a wide EGMS study corridor in the Lucanian sector of the Southern Apennines (southern Italy). The proposed approach combines a fully end-to-end deep learning pipeline tailored to the intrinsic characteristics of EGMS data, including strong non-stationarity, measurement noise, and frequent temporal gaps. To address these challenges, the framework integrates causal and robust preprocessing steps such as train-only normalization, local detrending, deterministic gap handling, and the incorporation of slope-derived auxiliary features. Building on this representation, we introduce a hybrid CNN–BiLSTM–Attention architecture that jointly captures short-term temporal fluctuations, longer-range dependencies within

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fixed historical windows, and the dynamic relevance of individual temporal lags. The forecasting task is evaluated through a spatially comprehensive validation strategy involving hundreds of thousands of EGMS measurement points and a strictly temporally disjoint test period, enabling a robust assessment of generalization across heterogeneous kinematic regimes and over the entire landslide body. Beyond predictive performance, the framework explicitly addresses the need for interpretability in geoscientific applications. Established explainable AI techniques are applied to analyse the temporal dependencies learned by the model, providing physically meaningful insights into the deformation dynamics of slow-moving landslides and linking deep-learning outputs with geomorphological understanding. Overall, this study shows that hybrid soft-computing methods can support short-term forecasting from continental-scale InSAR time series, with potential relevance for near-term monitoring and hazard assessment in slow-moving landslides.

2. Study Area and Dataset

2.1. Study Area and EGMS Product

The analysis focuses on satellite-derived ground deformation in southern Italy, a region characterized by complex topography and known landslide activity. The deformation data are obtained from the European Ground Motion Service (EGMS) Level 2b product, which provides time-series of ground motion derived from Sentinel-1 SAR acquisitions processed with multi-temporal InSAR techniques. The specific dataset used in this study is an EGMS L2b LOS deformation time series file (EGMS_L2b_044_0234_IW3_VV_2019_2023_1.csv), which contains cumulative LOS displacement values relative to a reference

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epoch, expressed in millimeters, for a dense set of measurement points within the selected study area (EGMS corridor / Area of Interest (AOI)). The regional corridor defined by the coordinate bounds reported in Section 2.2 is located within the Lucanian sector of the Southern Apennines, an area where clay-rich and structurally complex formations are widely recognized for their susceptibility to long-term ground deformation and slow-moving slope instabilities [28, 29, 30, 31]. In the Southern Apennines, clay-rich units—such as overconsolidated clays, varicoloured mudstones, and chaotic sedimentary complexes—are widely recognized for their tendency to undergo long-term, slow deformation, especially in slopes affected by deep-seated instability [32]. This creeping behaviour has been documented in different parts of the region, although the exact mechanisms and rates vary from site to site [33, 34, 35]. Independent research has also shown that slow-moving landslides in the Apennine chain commonly display seasonal to multi-annual variations in displacement rates driven by hydrological forcing. Typical patterns include accelerations during wetter periods—when increased pore-water pressures reduce shear strength—and decelerations during drier intervals, superimposed on the long-term creep component [36]. These characteristics are well documented across clay-dominated slopes in the Southern Apennines and provide the broader scientific context for interpreting the EGMS time series used in this study. No site-specific geological mapping or unit-level assumptions have been introduced; the description relies exclusively on established findings from peer-reviewed literature regarding clay-rich Apennine terrains.

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2.2. Number of Points and Spatial Sampling

The EGMS file provides deformation time series for more than 590,000 measurement points (594,978 EGMS ground-motion points). Each point corresponds to a persistent or distributed scatterer identified by the EGMS processing chain and is associated with geographic coordinates:

Latitude: 39.998756° N to 40.272107° N;

Longitude: 15.449297° E to 16.398026° E

These coordinates are used to generate spatial maps of deformation, velocity, forecasted values, and error metrics. The large number of points enables a spatially detailed characterization of ground deformation patterns over the study area but also imposes constraints in terms of memory usage and computational efficiency.

2.3. Temporal Coverage and Sampling

The time dimension is represented by a set of columns whose names follow the YYYYMMDD format (e.g. 20191010, 20191022), each corresponding to a Sentinel-1 acquisition date included in the EGMS product. These columns are automatically detected by matching 8-digit date strings and sorted chronologically to form a consistent temporal index. Let T denote the total number of acquisition dates. The dataset spans multiple years, from 2019 to 2023. For modelling, the temporal index is defined on a fixed 12-day cadence across the full period, so consecutive time steps correspond to a consistent 12-day forecasting horizon. Any missing acquisitions appear as

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gaps and are handled through the causal imputation described in Section 3.2. This temporal structure allows the construction of lagged input windows and corresponding prediction targets for data-driven time-series forecasting.

2.4. Signal Type: Cumulative Deformation Time Series

For each measurement point, the EGMS L2b product provides a cumulative line-of-sight (LOS) deformation time series representing the displacement relative to a reference epoch. In this study, the deformation signal for each point i is modelled as a cumulative displacement time series $x_i(t)$ expressed in millimetres, where t denotes the acquisition time index. The forecasting task is formulated as a next-step deformation prediction problem using a sliding temporal window. Specifically, for each point, the model input consists of the previous $L = 24$ observations (LOOKBACK = 24), while the prediction target is defined as the next-step deformation increment (delta mode), corresponding to the difference between consecutive cumulative deformation values in normalized space and subsequently denormalized to physical units. Under the fixed 12-day temporal cadence adopted in this study, the next-step prediction therefore corresponds to a consistent 12-day-ahead forecasting horizon.

2.5. Lag Structure and Feature Construction

The raw cumulative deformation time series are first normalized independently for each point using statistics computed exclusively from the training portion of the temporal axis, in order to prevent data leakage. For each point, the mean and standard deviation are estimated from the training time steps and subsequently used to standardize the entire series. From the normalized

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signals, fixed-length lagged windows of length $L = 24$ are extracted using a sliding-window approach. For each window, a multi-channel representation is constructed to encode complementary aspects of the deformation dynamics. The first channel corresponds to a locally detrended deformation signal, obtained by fitting a least-squares linear trend over the 24-point window and subtracting it to isolate short-term deviations from the local trend. The second channel encodes slope information by computing first-order temporal differences of the detrended signal within the same window, providing an explicit representation of short-term inter-epoch deformation increments. These two channels are stacked to form an input tensor of shape (L, C) with $L = 24$ and $C = 2$. To improve robustness against extreme values and rare artifacts, the resulting inputs are winsorized in normalized space to the range $[-6, 6]$. The prediction target for each sample is defined in delta mode as the normalized difference between the current and previous cumulative deformation values, which is subsequently converted back to physical units (millimetres) during evaluation and visualization.

2.6. Real-World Data Issues: Noise, Gaps, and Outliers

The EGMS time series exhibit several real-world imperfections that must be addressed before and during model training:

Missing values(gaps). Individual points may lack valid measurements at certain dates due to decorrelation, low coherence, or processing issues. Missing values are handled using a causal and deterministic imputation applied independently to each point-wise series. We use forward-fill only along the temporal axis, so that any missing entry is replaced by the most recent

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available observation and no future information can enter earlier training windows. Any remaining leading gaps (i.e., missing values before the first valid observation of a point) are filled with a fixed reference value (0 in our implementation) to guarantee a fully defined series prior to normalization and lag-window extraction.

Measurement noise and atmospheric artifacts. InSAR-derived deformation is affected by atmospheric phase delays, residual orbital errors, and decorrelation noise. These effects introduce high-frequency variability that is not directly related to true ground motion. The use of detrending, slope channels, and winsorization helps suppress non-physical fluctuations and stabilize learning.

Outliers and rare spikes. Some time steps may exhibit anomalously large apparent displacements due to unmodelled noise or local artefacts. To mitigate their impact on training, target values within each mini-batch are clipped using a robust bound defined as $\pm \max(0.5,\, 3\,\sigma_{\text{batch}})$, where σ_{batch} denotes the standard deviation of the target values in the current mini-batch (computed in normalized units). This prevents a small number of extreme samples from dominating the loss function.

Non-uniform temporal statistics. The variance of deformation across all points changes over time, reflecting the evolving dynamics of the slope and external forcing. Temporal variance and correlation analyses are computed to characterize these effects and to better interpret the forecasting results and model behavior. Through this combination of preprocessing steps and feature engineering, the raw EGMS deformation data are transformed into a clean,

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standardized, and physically meaningful input–target structure suitable for training and evaluating the hybrid deep learning model.

3. Methodology

3.1. Framework overview

This study proposes a data-driven forecasting framework designed to model cumulative, noisy, and non-stationary deformation time series derived from the EGMS product. The methodology is developed to operate at large spatial scale while preserving temporal causality and physical interpretability of deformation signals. The overall pipeline is organized into four sequential stages, comprising signal conditioning and normalization of cumulative deformation time series (1), the construction of fixed-length temporal windows with engineered features (2), deformation forecasting using a hybrid CNN–BiLSTM–Attention model (3) and the evaluation of predicted deformation which increments over a temporally disjoint interval (4). A central design choice is the formulation of the forecasting task in delta mode, where the model predicts short-term deformation increments rather than cumulative displacement values. This formulation directly addresses drift accumulation in cumulative time series and improves numerical stability during learning. All steps are applied consistently to the full EGMS dataset, enabling scalable forecasting across hundreds of thousands of spatial measurement points.

3.2. Signal conditioning and feature construction

The EGMS cumulative LOS deformation time series are affected by heterogeneous amplitudes, missing observations (gaps), and occasional non-physical

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artefacts. In this study, the time axis is defined on a fixed 12-day cadence, yielding a consistent one-step forecasting horizon. To ensure stable learning while preserving the temporal structure of the signals, each time series is processed independently using a fixed and deterministic sequence of operations.

Missing observations (gaps) are resolved through a point-wise imputation applied along the temporal axis. Specifically, missing entries are completed using forward-fill only, yielding a causally imputed series prior to normalization and lag-window extraction. Forward-fill preserves temporal causality by ensuring that each imputed value depends only on previously observed acquisitions; no backward-fill is used for model inputs during training/validation/testing, and backward-fill is applied only in the exploratory time-correlation diagnostic (Fig. 3) to obtain a complete correlation matrix for visualization. Any remaining leading gaps (before the first valid measurement of a point) are assigned a constant reference value (0 in our implementation) to keep the series fully defined for downstream processing. For standardization, mean and standard deviation are computed per point using only the training portion of the time axis and ignoring missing values (i.e., computed on the available observations after forward-fill). These train-only statistics are then applied to normalize the full series for that point, reducing temporal leakage risk and stabilizing optimization.

After normalization, fixed-length temporal windows are extracted using a sliding-window approach. Each window spans $L = 24$ acquisition epochs and represents the historical context used to forecast the next deformation increment. Within each window, a least-squares linear trend is estimated and removed, yielding a locally detrended signal that suppresses low-frequency drift

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over the context interval while highlighting short-term deviations. To explicitly encode short-term kinematic information, first-order temporal differences of the detrended signal are computed within each window and stacked as an additional channel. The final input is therefore represented as a two-channel tensor composed of the detrended normalized deformation and its corresponding slope signal.

To improve robustness against atmospheric artefacts and decorrelation-induced anomalies, input features are winsorized to the range $[-6, 6]$ in normalized units. In addition, to mitigate the influence of rare spikes during optimization, target values are clipped at the mini-batch level using a robust threshold proportional to the batch standard deviation.

3.3. Delta-based forecasting formulation

The forecasting task is formulated as a next-step prediction problem operating in delta mode. Let $x_i(t)$ denote the cumulative LOS deformation of point i at time t, and $\tilde{x}_i(t)$ its normalized representation. The target variable is defined as the inter-epoch deformation increment [37, 38] :

$$y_i(t) = \tilde{x}_i(t) - \tilde{x}_i(t-1) \tag{1}$$

Under this formulation, the model learns to predict short-term deformation changes conditioned on the previous 24 observations. Predicting increments rather than cumulative values mitigates drift accumulation and enables consistent learning across points characterized by different absolute deformation magnitudes.

During inference, predicted normalized increments are converted back to physical units using the standard deviation estimated from the training

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data. When cumulative deformation values are required for visualization or interpretation, they are reconstructed by adding the predicted increment to the last observed displacement. Quantitative evaluation is reported on the denormalized one-step-ahead deformation obtained after reconstruction (i.e., adding the denormalized increment to the last observed displacement). Because both prediction and reference share the same last-observed baseline, the resulting errors are numerically equivalent to evaluating the denormalized deformation increments.

3.4. Hybrid CNN–BiLSTM–Attention model

To capture both local and long-range temporal dependencies within deformation windows, a hybrid neural architecture combining convolutional, recurrent, and attention-based components is adopted.

Each input sample is represented as a two-channel temporal sequence of length 24. Local temporal patterns and short-term fluctuations are first extracted using a causal one-dimensional convolutional front-end. These convolutional layers operate along the temporal dimension and act as a feature extractor that enhances informative local structures while attenuating high-frequency noise. Causal padding ensures that each temporal representation depends only on present and past observations.

The convolutional features are then processed by stacked BiLSTM layers. The BiLSTM backbone captures longer-range temporal dependencies by processing the windowed sequence in both forward and backward directions, allowing the model to encode evolving deformation behaviour within each input window. Sequence-level outputs are preserved to support subsequent temporal weighting.

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Although Bidirectional LSTM layers process the input sequence in both forward and backward directions, their use in this framework does not violate temporal causality. The bidirectional processing is strictly confined to a fixed-length historical window composed exclusively of past observations relative to the prediction time. No information from future acquisition epochs beyond the forecast horizon is accessed during training or inference. The BiLSTM architecture is therefore employed solely to enhance the internal representation of temporal dependencies within the available historical context, improving feature extraction without introducing temporal leakage.

A multi-head self-attention mechanism is applied to the BiLSTM outputs to dynamically weight the contribution of individual time steps. This attention block enables the model to emphasize the most informative temporal lags before aggregation, improving both predictive performance and interpretability of temporal relevance. The attention-weighted sequence is subsequently condensed into a fixed-length context vector and passed to a fully connected regression head that outputs the predicted next-step deformation increment.

3.5. Training strategy and evaluation protocol

Model training follows a supervised learning paradigm with a strict chronological separation between training, validation, and test intervals to preserve temporal causality. For each point-wise deformation time series, the first 70% of available acquisitions are used for training, the subsequent 15% for validation, and the remaining 15% for testing. Lagged input windows are defined as the $L = 24$ observations immediately preceding each target time step. Prediction targets are sampled within each chronological subset (train/validation/test); however, windows located near split boundaries may

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include past observations from earlier subsets, while never accessing any observation beyond the prediction epoch.

Mini-batch sampling is performed probabilistically over target time indices within each chronological subset using a non-uniform temporal sampling scheme (implemented via a power-transformed uniform draw with `BIAS_POWER`= 2.0 in our implementation). Spatial points are sampled uniformly. Robust regression loss functions are employed to accommodate occasional noisy deformation increments, and batch-level clipping of extreme target values further limits the influence of anomalous samples during training.

Model evaluation is conducted on a temporally held-out test interval. The network outputs the next-step deformation increment in normalized units; for reporting, predictions are denormalized using the per-point training standard deviation and reconstructed into denormalized one-step-ahead cumulative deformation by adding the denormalized increment to the last observed displacement within each input window. Performance is assessed using mean absolute error (MAE), root mean squared error (RMSE), and coefficient of determination (R^2) computed on this reconstructed one-step-ahead deformation.Because the number of eligible (point·time) pairs is very large at regional scale, metrics are computed on a fixed, reproducible random sample of N_{test} windows (with $N = \max(200{,}000,\ N_{\text{XAI}})$ and $N_{\text{XAI}} = 3{,}000$ in our implementation), with spatial points sampled uniformly and time steps drawn using the same temporal sampling procedure used during training. The same fixed test sample is also reused for explainability analyses (Section 4.7) to ensure consistent attribution comparisons.

Post-hoc explainability analyses are subsequently employed to interpret

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the temporal contributions learned by the model, and are described in Section 4.7. Baseline models (including a gated recurrent unit (GRU) forecaster) are trained and evaluated under the same preprocessing, temporal split, input configuration (`LOOKBACK` $= L = 24$ with the same channels), and reconstruction/denormalization pipeline to ensure a fair comparison.

3.6. Summary of the methodological pipeline

For clarity, Table 1 summarizes the methodological pipeline and the role of each processing and modelling component. Tables 2 and 3 report the main architectural and training hyperparameters of the CNN–BiLSTM–Attention model, providing a concise and reproducible specification of the network design.

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Table 1: Summary of the proposed forecasting methodology

Stage	**Component**	**Description**
Data normalization	Per-point standardization	Training-based standardization of cumulative deformation.
Signal conditioning	Local detrending	Local linear trend removal within temporal windows.
Feature encoding	Slope channel	First-order differences encoding deformation increments.
Robustification	Outlier handling	Input winsorization and robust mini-batch target clipping.
Input construction	Lagged windows	Two-channel tensors from fixed 24-epoch windows.
Forecasting task	Delta-mode prediction	Prediction of deformation increments instead of displacement.
Local pattern extraction	1D convolutional layers	Causal convolutions for local temporal pattern extraction.
Temporal dependency modelling	Bidirectional LSTM	BiLSTM layers modelling bidirectional temporal dependencies.
Temporal weighting	Self-attention mechanism	Multi-head attention weighting informative temporal lags.
Output layer	Regression head	Regression head outputting normalized deformation increments.

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Table 2: CNN–BiLSTM–Attention architecture: data and network components

Component	**Specification**
Input window length	24 acquisition epochs
Input channels	2 (detrended normalized deformation, slope)
Input pre-processing	Per-window detrending, train-only standardization, winsorization to $[-6, 6]$
Target formulation	Next-step deformation increment ($\Delta x(t)$)
Target pre-processing	Robust batch-level label clipping
Conv1D layer 1	64 filters, kernel = 3, causal, GELU
Conv1D layer 2	64 filters, kernel = 5, dilation = 2, causal, GELU
CNN regularization	Dropout = 0.25
BiLSTM layer 1	128 units, bidirectional, return sequences
BiLSTM layer 2	64 units, bidirectional, return sequences
Recurrent regularization	Dropout = 0.25, recurrent dropout = 0.1
Normalization	Layer normalization after BiLSTM and attention
Attention mechanism	Multi-head self-attention with residual connection
Attention configuration	4 heads, key dimension = 64
Temporal aggregation	Global average pooling
Dense layer	128 units, GELU
Output layer	1 unit (deformation increment)

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Table 3: CNN–BiLSTM–Attention architecture: training and optimization

Component	**Specification**
Loss function	Huber loss ($\delta = 1.0$)
Optimizer	AdamW (weight decay $= 10^{-4}$, clipnorm $= 1.0$)
Learning-rate schedule	Cosine decay with restarts ($\mathrm{lr}_0 = 10^{-3}$)
Batch size	1024
Training epochs	60
Early stopping	Patience = 10 (validation loss)
Training strategy	Fixed steps per epoch (600 train / 120 validation)
Sampling strategy	Non-uniform temporal target sampling (power-law, $p = 2.0$)

The incremental target is computed in normalized space and subsequently denormalized for reporting in physical units (e.g., mm). Table 2 adopts Δx as a shorthand notation. To stabilize training under occasional non-physical artefacts, we apply two lightweight robustification steps. First, normalized input windows (after detrending and optional slope-channel construction) are winsorized to a fixed range (here $[-6, 6]$) to limit the influence of extreme outliers. Second, during mini-batch generation we apply a robust label clipping rule, constraining the incremental target within $\pm k\sigma$ (with $k = 3$) computed on the batch, with a small lower bound to avoid numerical issues. Importantly, this label clipping is applied only during training/validation batch sampling and is not used when computing test metrics.

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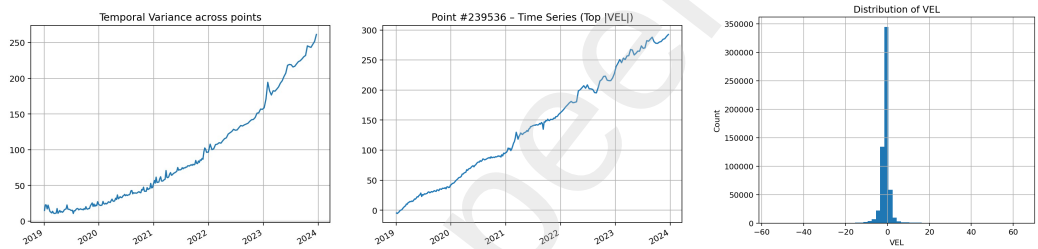
4. Results

4.1. Exploratory Analysis of the Satellite Deformation Time Series

Temporal investigation of the cumulative displacement revealed a persistent increase from 2019 onward, with a progressive widening of temporal variability (Fig. 1a). Sample inspection of highly deforming points confirmed non-linear trends, intermittent local accelerations, and stabilization phases (Fig. 1b). Moreover, the histogram of velocities revealed a positively skewed distribution toward higher deformation rates (Fig. 1c).



(a) Temporal variance across spatial points

(b) Representative time series from a high-velocity point

(c) Distribution of deformation velocities

Figure 1: Exploratory analysis of deformation variability and velocity characteristics.

4.2. Spatial Distribution of Deformation and Velocity

The spatial map of cumulative deformation at the final observation step highlighted a distinct concentration of displacement in the central–southeastern sectors of the study area (Fig. 2a). These same areas exhibited the highest deformation rates (Fig. 2b), confirming that long-term cumulative deformation is strongly associated with persistent kinematic acceleration.

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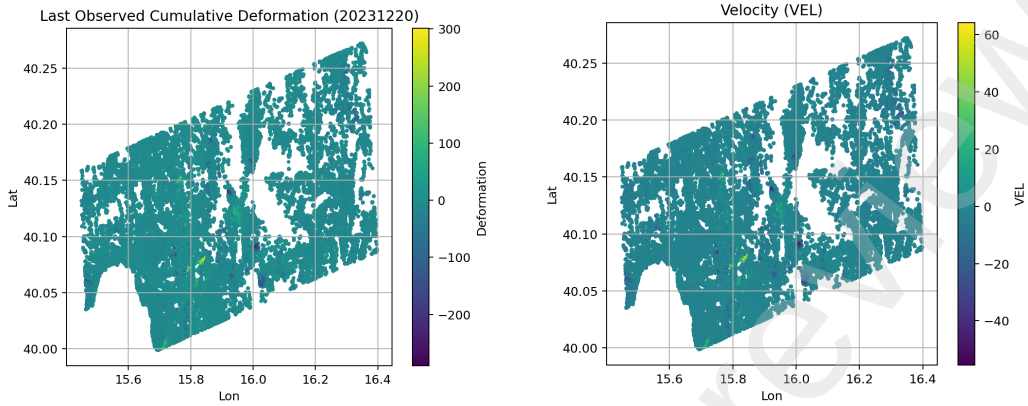

(a) Last observed cumulative deformation

(b) Spatial distribution of deformation velocity

Figure 2: Spatial patterns of cumulative deformation and deformation velocity.

In addition to the spatial patterns, a time-step correlation diagnostic was computed across acquisition epochs (Fig. 3). The matrix represents correlations between acquisition dates computed from the cumulative deformation values across points (using a subset of time steps for visualization). Correlations are strongest for temporally adjacent epochs, which is expected for cumulative displacement signals that evolve smoothly over time. For this diagnostic visualization, missing values in the selected time-step subset are deterministically filled (forward/back-filled) prior to correlation computation to obtain a complete correlation matrix.This completion step is used exclusively for the correlation visualization and does not affect the causal preprocessing used to construct training and evaluation windows.

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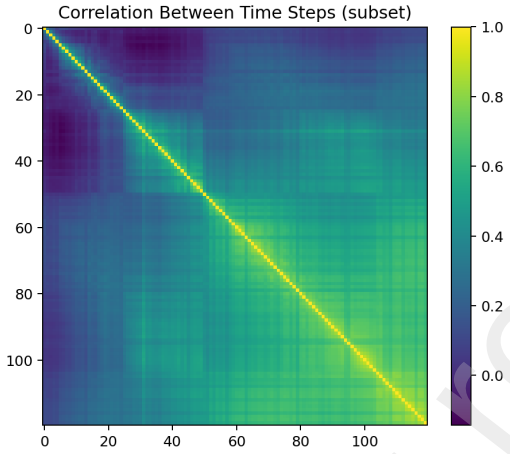

Figure 3: Correlation matrix between acquisition epochs computed from cumulative deformation values (subset of time steps).

Point-wise prediction errors remained low and spatially homogeneous (Fig. 4a), with no localized degradation zones, suggesting stable model generalization across the full study extent. Moreover, the predicted next-step deformation map closely reproduced the observed spatial gradients (Fig. 4b), including within areas characterized by higher deformation rates, supporting qualitative spatial consistency between observations and forecasts.

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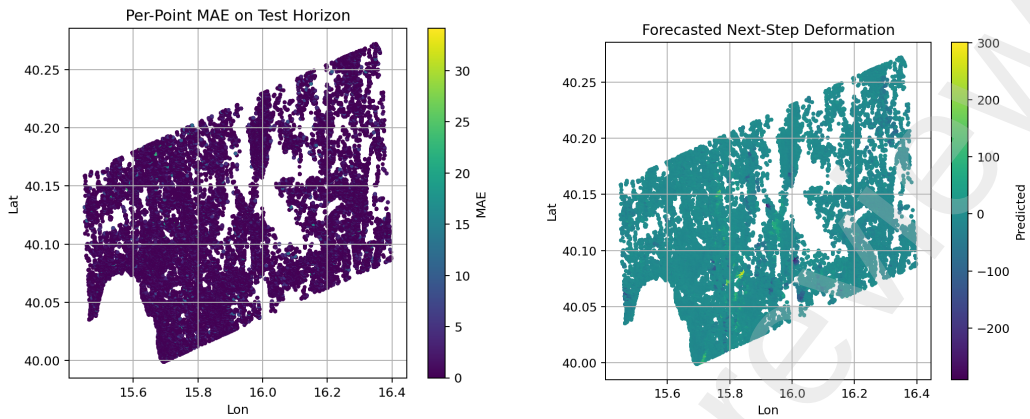

(a) Per-point MAE on the test horizon

(b) Predicted next-step deformation

Figure 4: Spatial patterns of forecasting error and next-step deformation.

4.3. Temporal Pattern Separation Using PCA and Clustering

To explore variability in the temporal evolution of deformation, trajectories were projected into a low-dimensional Principal Component Analysis (PCA) space (Fig. 5a). Prior to PCA, each point-wise cumulative deformation trajectory was standardized independently (zero mean and unit variance), so that the embedding primarily reflects differences in temporal profile shape rather than absolute deformation magnitude. For computational efficiency, PCA was fit on a random subset of up to 150,000 points.

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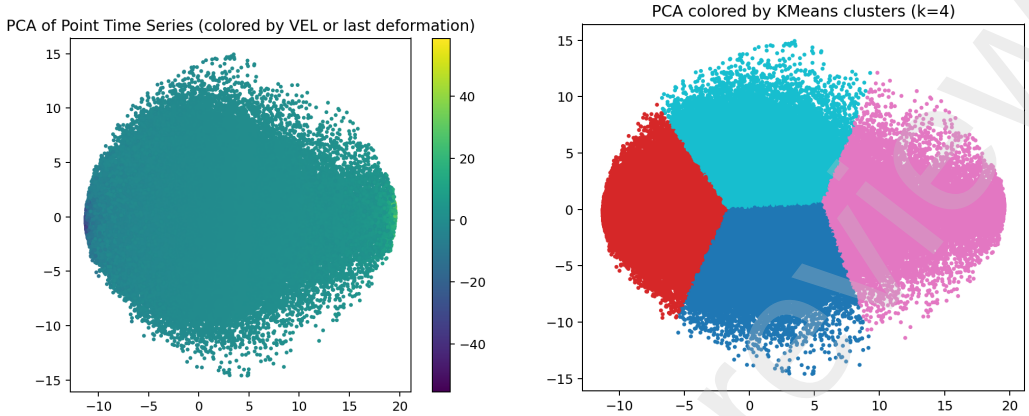

(a) PCA of point-wise deformation trajectories (colored by velocity or last deformation)

(b) Same PCA projection colored by KMeans clusters

Figure 5: Low-dimensional representation of deformation time series and cluster structure in PCA space.

Based on this embedding, KMeans clustering was applied to the same sampled points to partition dominant trajectory patterns (Fig. 5b). Cluster-wise mean trajectories (with interquartile ranges) were then computed in original deformation units for the points within each cluster (Fig. 6), providing representative deformation behaviours across the study area. The resulting groups summarize characteristic modes ranging from quasi-stable behaviour to progressively deforming and more oscillatory trajectories, supporting qualitative interpretation of heterogeneous temporal patterns in the EGMS deformation series.

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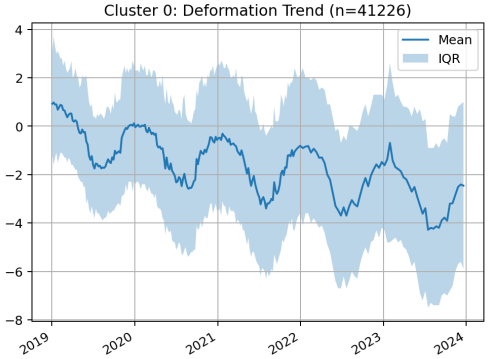
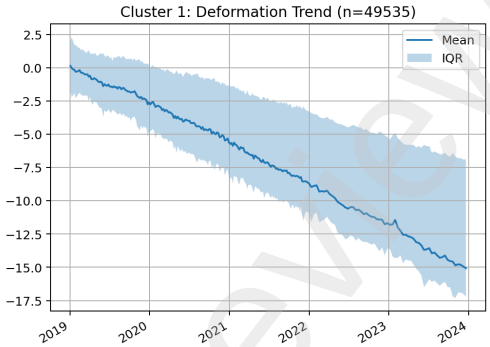
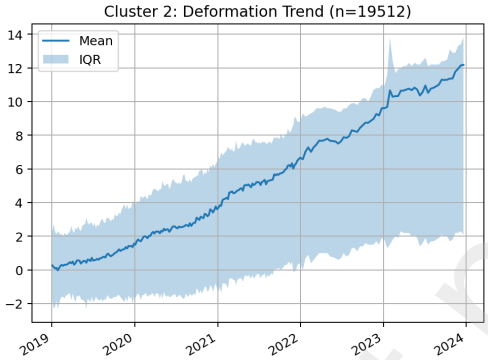


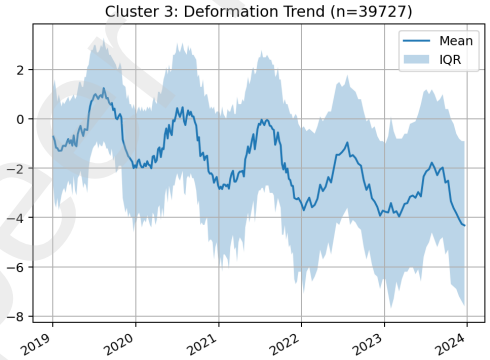
(a) Cluster 0



(b) Cluster 1



(c) Cluster 2



(d) Cluster 3

Figure 6: Representative deformation trends (mean and interquartile range) for the four clusters derived from PCA-space clustering (computed on the sampled points used for PCA/KMeans).

These cluster-specific temporal signatures qualitatively align with the spatial patterns discussed in the following sections, where regions assigned to progressively deforming clusters tend to coincide with areas of higher velocity and stronger short-term temporal dependence.

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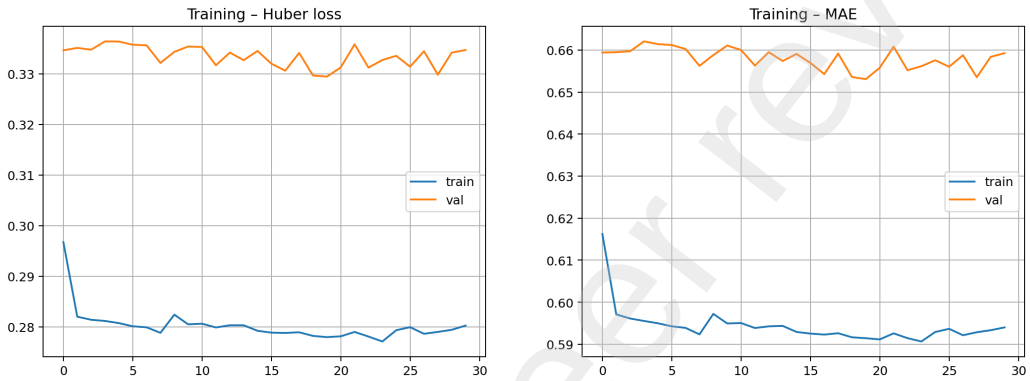
4.4. Model Training Behavior

The evolution of the training loss and validation MAE indicated stable convergence, with both curves decreasing smoothly and reaching near-plateau levels after approximately 15 epochs (Fig. 7).



(a) Training Huber loss

(b) Training MAE

Figure 7: Training convergence behaviour of the proposed model.

The absence of divergence between training and validation metrics demonstrates good generalization performance and suggests that the model did not exhibit overfitting during optimization.

4.5. Baseline comparisons

To contextualize the proposed CNN–BiLSTM–Attention architecture, we report a standard recurrent baseline based on a GRU forecaster trained under the same preprocessing pipeline, the same input configuration (`LOOKBACK` $= L = 24$ with the same two channels), the same delta-mode target definition, and the same temporally held-out evaluation protocol. In addition, we reporta persistence (naïve) baseline defined as $\hat{x}_{t+1} = x_t$ (equivalently, $\widehat{\Delta x}_{t+1} = 0$

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in delta mode), which provides a non-learning reference under the same reconstruction/denormalization procedure.

On the test interval, the proposed model achieves MAE = 2.925 mm, RMSE = 4.237 mm, and $R^2 = 0.903$, outperforming both the GRU baseline (MAE = 3.267 mm, RMSE = 4.886 mm, $R^2 = 0.840$) and the persistence baseline (MAE = 5.234 mm, RMSE = 7.521 mm, $R^2 = 0.791$) (Table 4).

Relative to the GRU baseline, this corresponds to a reduction of 0.342 mm (10.47%) in MAE and 0.649 mm (13.28%) in RMSE, with an R^2 increase of 0.063. Relative to persistence, the proposed model reduces MAE by 2.309 mm (44.12%) and RMSE by 3.284 mm (43.66%), with an R^2 increase of 0.112. The GRU baseline itself improves over persistence (MAE reduction of 1.967 mm (37.58%) and RMSE reduction of 2.635 mm (35.04%)), indicating that learned recurrent forecasting provides measurable benefit beyond simply repeating the last observation (Table 4).

Table 4: Baseline comparison on the held-out test interval.

Model	MAE (mm)	RMSE (mm)	R^2
Proposed CNN–BiLSTM–Attention	2.925	4.237	0.903
GRU baseline	3.267	4.886	0.840
Naïve (Persistence)	5.234	7.521	0.791

4.6. Forecasting Performance

Test-set evaluation demonstrated high accuracy, supported by the tight alignment between predicted and observed denormalized one-step-ahead deformation (reconstructed from the predicted increment) (Fig. 8a). The

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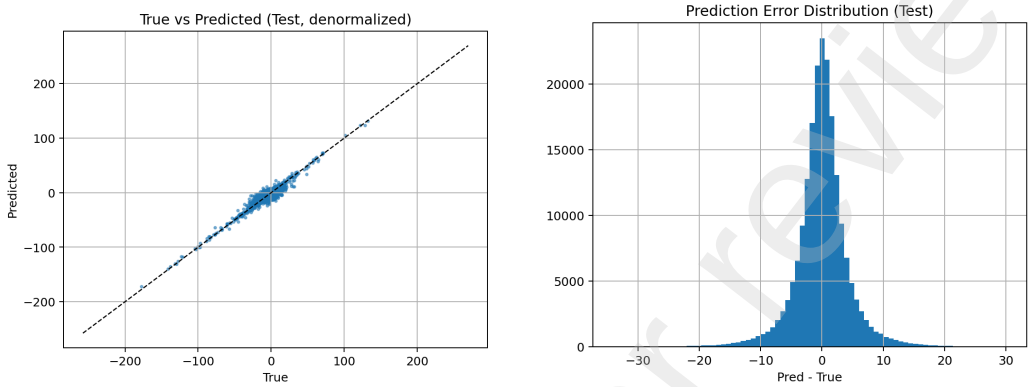
distribution of prediction errors was centered around zero, with the majority of absolute deviations remaining below 10 mm (Fig. 8b).



(a) True vs. predicted one-step deformation

(b) Prediction error distribution

Figure 8: Test-set one-step prediction accuracy and error patterns.

Additionally, the worst-case example maintained consistency with the corresponding deformation evolution (Fig. 9), indicating that the forecasting model remains stable even under accelerated deformation scenarios.

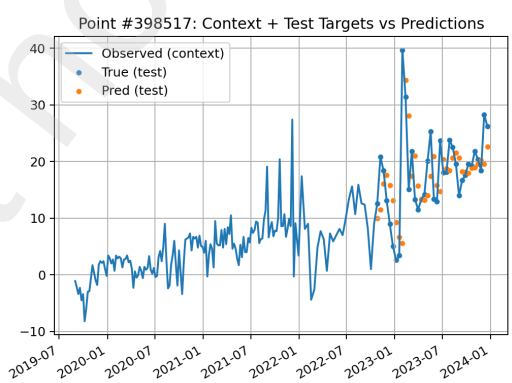


Figure 9: worst point timeline

Quantitatively, the forecasting model achieved MAE = 2.925 mm, RMSE =

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4.237 mm, and $R^2 = 0.903$ on the temporally held-out test interval (Table 4). These values indicate that most one-step-ahead (12-day-ahead) prediction errors remain within a few millimetres. Considering the noise level and acquisition-related uncertainty inherent to EGMS InSAR measurements, this level of accuracy is consistent with the expected precision of satellite-derived deformation products.

The high coefficient of determination further shows that a large fraction of the variance in the denormalized one-step-ahead deformation is explained by the model, supporting robust predictive performance across heterogeneous kinematic regimes.Because a large fraction of test samples exhibit near-zero increments, overall metrics can be dominated by stable regimes. We therefore additionally report performance on an active subset (top 10% of test samples ranked by $|y|$) to quantify accuracy under the most dynamic deformation conditions (Table 5). As expected, errors increase on the active subset; however, performance remains robust with R^2 staying above 0.82, indicating stable predictive skill under high-activity regimes.

Table 5: Stratified test performance of the proposed model on the temporally held-out interval (overall vs. active subset). y denotes the observed one-step (12-day) deformation increment in mm. The active subset includes the top 10% of test samples ranked by $|y|$ (i.e., $|y| \geq$ 90th percentile).

Evaluation subset	Definition	MAE (mm)	RMSE (mm)	R^2
Overall test	All test samples	2.925	4.237	0.903
Active subset	Top 10% by $\|y\|$	3.539	5.381	0.821

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4.7. Explainability of Deformation Forecasting

Explainability is assessed using gradient-based attribution methods to quantify the temporal contribution of individual lags within each input window. Integrated Gradients (IG) is employed as the primary attribution method, using a zero baseline corresponding to the absence of deformation signal in the normalized input space [39] . To improve attribution stability, Smooth Integrated Gradients is adopted by averaging IG scores over multiple noisy realizations of the input sequence, generated through small Gaussian perturbations applied to the normalized features [40, 41] .For reproducibility, Integrated Gradients are computed with 64 interpolation steps and a zero baseline. To reduce attribution variance, we additionally report Smooth Integrated Gradients, obtained by averaging IG over 6 noisy realizations of each input window using Gaussian noise with standard deviation 0.05 (in normalized units) and 32 IG steps per realization. Attribution analyses are computed on a fixed subset of 3,000 test windows, consistent with the sampling protocol described in Section 3.5.

Attribution faithfulness is evaluated through deletion and insertion tests [42] . In the deletion setting, input time steps are progressively removed from each temporal window in descending order of attribution magnitude, and the resulting degradation in predictive performance is measured. Conversely, the insertion test progressively restores the most relevant time steps starting from a baseline input, quantifying the corresponding improvement in prediction accuracy. These complementary analyses provide an objective assessment of whether the attribution rankings reflect the true predictive dependencies of the model.

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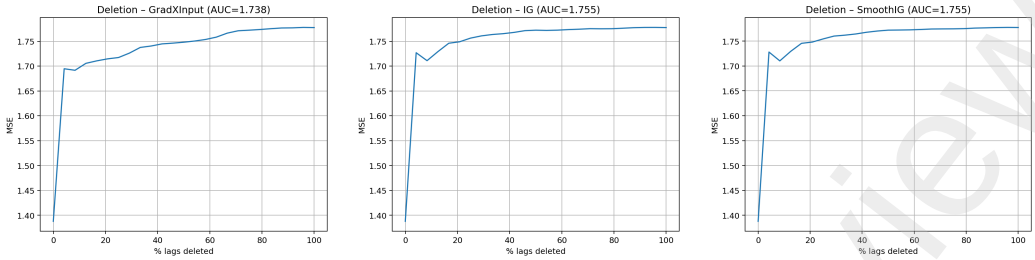
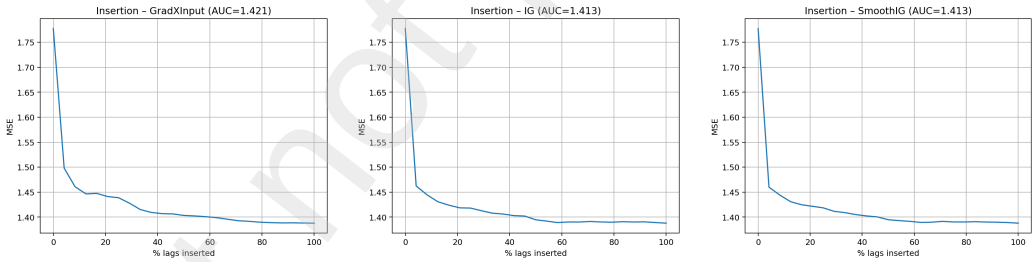


(a) Grad×Input (b) Integrated Gradients (c) Smooth Integrated Gradients

Figure 10: Deletion-based faithfulness analysis across attribution methods.

Deletion-based analyses demonstrate that removing the most highly ranked recent lags leads to a rapid increase in prediction error across all attribution methods (Fig. 10), indicating strong reliance on near-term temporal information. Conversely, insertion curves confirm that reintroducing these same lags produces the largest performance gains (Fig. 11), providing evidence that the attribution rankings are faithful to the model's predictive behaviour.



(a) Grad×Input (b) Integrated Gradients (c) Smooth Integrated Gradients

Figure 11: Insertion-based faithfulness analysis across attribution methods.

Consistently across all global attribution methods (Fig. 12b), the most recent time step emerges as the dominant contributor, followed by the immediately preceding lags, while more distant history exhibits markedly lower importance. This temporal relevance pattern aligns with the incremental

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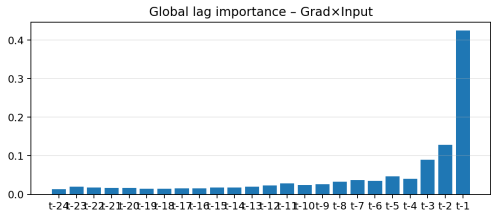
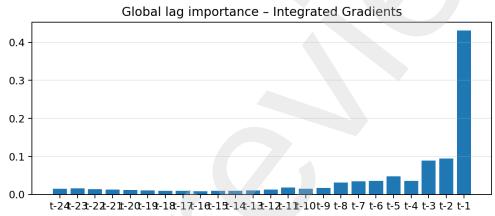
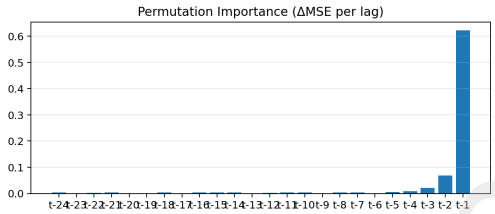
forecasting formulation and reflects the physical expectation that short-term deformation variations exert the strongest influence on near-term displacement evolution.



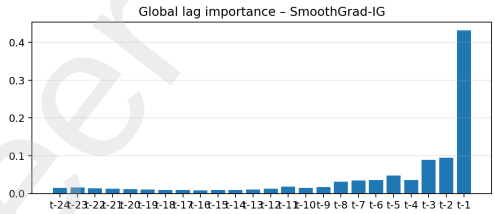
(a) Global lag importance – Grad×Input



(b) Global lag importance – Integrated Gradients



(c) Permutation-based lag importance



(d) Global lag importance – Smooth Integrated Gradients

Figure 12: Global temporal lag importance estimated using different explainability methods.

From a geophysical perspective, the dominance of recent temporal lags is consistent with the behaviour of slow-moving landslides, where short-term deformation variations are often driven by transient hydrological forcing, pore-pressure changes, and near-surface mechanical adjustments rather than long-term cumulative displacement alone [43]. Seasonal recharge and drainage cycles, commonly reported in clay-rich Apennine slopes, tend to modulate deformation rates at short temporal scales, making recent observations more informative for near-term forecasting [44, 45]. The observed attribution patterns therefore support not only the internal consistency of the model, but also its alignment with physically plausible deformation mechanisms.

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4.8. Spatial Explainability Patterns

The spatial attribution surface (Fig. 13) revealed localized patches with stronger reliance on recent temporal information. When compared with the velocity field (Fig. 2b), these high-attribution areas co-located with zones of higher deformation rate, suggesting consistency between the physical deformation evolution and the model's learned temporal relevance.

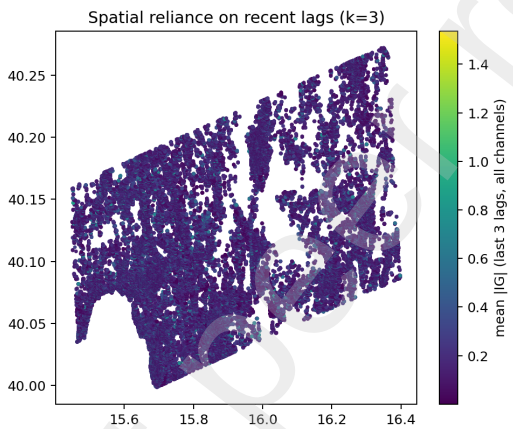

Figure 13: Spatial reliance on recent lags

To relate these attribution patterns to monitoring-relevant regimes, we further connect the explainability results with activity-stratified performance. While the dominance of recent lags in one-step incremental forecasting is expected, we do not interpret the attribution profiles as a purely trivial recency effect. The Smooth Integrated Gradients analysis (Fig. 12) quantifies how much the model relies on short-term history, and the spatial attribution map (Fig. 13) shows that this reliance is not uniform across the study area, but concentrates in localized regions characterized by more variable deformation behaviour.

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Importantly, the activity-stratified evaluation (Table 5) confirms that the model retains substantial predictive skill on the most dynamic cases (top 10% by $|y|$), with $R^2 = 0.821$ despite the expected increase in MAE/RMSE. Taken together, these results indicate that the learned short-memory dependence is not merely driven by near-zero increments, but remains informative for monitoring-relevant high-activity regimes.

5. Discussion

The results obtained in this work demonstrate that the adopted forecasting strategy is well aligned with the intrinsic characteristics of EGMS displacement time series and provides stable short-term predictions over a large spatial extent. The use of temporally disjoint partitions ensures that the evaluation reflects genuine forecasting ability under a chronological train/validation/test split, and the observed performance on the final unseen interval suggests that deformation progression was adequately captured. To obtain complete point-wise sequences prior to normalization, missing observations are deterministically completed using forward-fill only along the time axis. Because this imputation is strictly causal, it does not propagate information from future acquisitions into earlier missing entries. Any leading gaps are filled with a fixed reference value (0 in our implementation) so that normalization and window extraction remain well-defined for all points. As with any gap-filling strategy, short missing intervals may be smoothed, which is considered when interpreting predictive performance. This aspect is essential when working with slow-moving landslides, as deformation trajectories often exhibit phase shifts, non-monotonic behaviour, or secondary

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variations unrelated to early-stage dynamics. The formulation of the learning objective in incremental form (delta mode) played a central role in improving prediction consistency. Since cumulative displacement increases over long periods, direct learning of absolute levels would have forced the model to reproduce long-range drifts partly controlled by time-dependent referencing and atmospheric filtering processes. By focusing on inter-epoch changes, the model primarily relied on deformation variations that are more representative of the current slope state rather than historical accumulation. The improved numerical stability observed during training suggests that this formulation reduced the influence of long-term offset trends and enabled clearer separation between background creep behaviour and short-lived adjustments. The incorporation of detrended signals and a slope-based channel within each temporal window also contributed to model stability. The detrending step removes low-frequency drift within the window, preventing recurrent layers from encoding linear patterns with limited predictive value at short horizons. Meanwhile, the slope channel highlights local increment fluctuations that correspond more directly to deformation evolution. This separation allowed the learning process to emphasize inflection behaviour rather than cumulative magnitude, which is particularly relevant where deformation changes are gradual and dominated by subtle accelerations rather than abrupt events. These preprocessing steps are consistent with the temporal nature of EGMS series, where high-frequency variations often result from non-ground-motion artefacts, whereas systematic increment variations convey physically interpretable behaviour [46]. The architecture employed in this study combines convolutional filtering, bidirectional recurrence, and adaptive temporal weighting.

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The convolutional layers attenuate isolated fluctuations while extracting local features prior to sequence modelling, which is advantageous given the presence of short-lived deviations in the signal. The bidirectional recurrent layers, although applied over a short window, provide access to deformation evolution within the contextual interval and help distinguish monotonic sequences from sequences characterized by alternating increases and stabilizations. The attention mechanism then reweights contributions within the window, emphasizing periods with stronger local variations. The relevance of recent lags indicated by the attribution analysis is consistent with the expected contribution of actual displacement increments observed at the end of the time sequence. These outcomes align with the known kinematic behaviour of slow-moving slopes, where short-duration variations frequently precondition the immediate response rather than long-term accumulation. The spatial distribution of prediction errors further reinforces this interpretation. Locations with highly variable temporal signatures frequently exhibited larger deviations between predicted and observed increments, whereas areas showing gradual monotonic deformation yielded more consistent forecast behaviour. This observation suggests that prediction uncertainty is more sensitive to temporal variability rather than spatial context per se. Given that irregularity is often associated with decorrelation, transient atmospheric components, or sparse valid observations, such results are expected in large-scale SAR-derived datasets and do not reflect a structural deficiency of the learning framework. In addition to the overall metrics, the activity-stratified evaluation (Table 5) indicates that predictive skill remains non-trivial under the most dynamic deformation conditions, despite the expected increase in errors. Additionally, several

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methodological components played an auxiliary but important role in stabilizing model behaviour, including label-wise robust clipping to mitigate rare atmospheric spikes, the use of a consistent probabilistic time-index sampling scheme for both training and evaluation, and the adoption of a two-channel lag formulation that indirectly enforces partial stationarity within short temporal windows. Taken together, the findings of this study indicate that meaningful short-term forecasting of slow-moving landslide deformation can be achieved when the learning formulation reflects both the acquisition-driven characteristics of satellite observations and the temporal nature of creep evolution. The interplay between detrending, increment-based supervision, noise-robust sampling, and adaptive temporal weighting results in prediction behaviour consistent with the physical interpretation of displacement development. The observations presented here demonstrate that model response is shaped not by purely statistical adaptation but by temporal relationships aligned with the deformation process itself, supporting the suitability of the proposed strategy for operational or extended monitoring settings.

6. Conclusions

This study introduced a scalable forecasting framework specifically tailored to deformation time series obtained from the EGMS product, demonstrating that short-term predictions of slow-moving landslides can be achieved reliably when the temporal characteristics of satellite-derived observations are embedded into the modelling strategy. The formulation of the task in incremental mode was central to this outcome, as it removed the influence of cumulative referencing effects and allowed the model to focus on deformation

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variations that are more representative of current kinematic conditions. This decision prevented the propagation of long-term drift into prediction targets and yielded numerically stable training behaviour.

The preprocessing pipeline adopted in this work contributed significantly to the robustness of the forecasting results. Local detrending within each temporal window separated slow baseline motion from short-term fluctuations, while slope-based channelization provided an explicit representation of inter-epoch changes. Together with winsorization and robust label clipping, these steps minimized the impact of non-physical measurement spikes, decorrelation artefacts, and temporal gaps often present in EGMS time series. This ensured that the model received input windows with meaningful deformation structure, rather than cumulative noise-affected trajectories. From a modelling standpoint, the proposed architecture combined multi-scale convolutional extraction, bidirectional recurrence, and adaptive temporal weighting. The convolutional component suppressed isolated fluctuations before temporal integration, while the BiLSTM layers captured deformation evolution within the recent window from both forward- and backward-looking temporal relationships. The attention mechanism further refined this representation by emphasizing the most relevant segments of the window. This prioritization was confirmed through attribution analysis, where the highest temporal relevance consistently corresponded to the most recent lags—aligning with the definition of a one-step-ahead prediction problem.

Validation over a temporally held-out period demonstrated stable generalization without degradation at locations undergoing higher deformation rates. Forecast accuracy remained spatially homogeneous despite strong variability

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in temporal profiles among individual measurement points. The error patterns were primarily driven by local irregularity in temporal behaviour rather than spatial clustering, indicating that the forecasting strategy is resilient to spatial heterogeneity and operationally applicable to large areas. This is particularly relevant for large-scale monitoring environments where reliable predictions must be obtained across thousands of independent time series.

A key novel contribution of this work lies in the integration of incremental supervision, local detrending, slope-encoded multichannel representation, and non-uniform temporal sampling of target windows within a unified deep learning pipeline. While individual components are known in time-series modelling, their combined formulation specifically for EGMS-derived deformation forecasting, and at a scale exceeding half a million observation points, has not been systematically reported. Moreover, embedding explainability directly onto the trained model provides insight into how displacement histories influence predictions, offering transparency that is often absent in black-box forecasting approaches.

Beyond the scope of the present experiments, an impactful extension would be a potentially impressive extension of this work, especially to address the noise, gaps, and operational stakes of EGMS forecasting, is to complement the current purely data-driven pipeline (delta-mode next-step prediction, window detrending + slope channel, and attention with IG-based temporal attributions) with a symbolic / neuro-symbolic "reliability layer" that formalizes and checks domain expectations about deformation dynamics. Concretely, monitoring-relevant requirements can be expressed as temporal specifications over real-valued signals—e.g., bounded acceleration/jerk, limits on implausible

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oscillations after detrending, regime-consistent behavior (progressive vs seasonal clusters), and conditions linking missingness to uncertainty—and then used (i) post hoc to verify neural forecasts and flag trajectories that violate physically or operationally meaningful constraints, or (ii) during training as constraint regularizers, where specification violations contribute an additional loss that discourages pathological forecasts in gap-heavy or highly irregular segments [47, 48]. This direction is technically feasible because Signal Temporal Logic (STL) and related monitoring frameworks support quantitative robustness notions for continuous signals, and these robustness measures have been widely used for optimization and control—providing a practical bridge toward differentiable, "logic-guided" learning and verification workflows [49]. More broadly, neuro-symbolic AI has been advocated as a principled way to combine the pattern-recognition strength of deep models with the explicitness, auditability, and guarantee-oriented reasoning of symbolic systems, which is particularly relevant in safety-critical monitoring pipelines where stakeholders require explanations and enforceable operating rules [50]. Finally, beyond pointwise forecasting, symbolic components could support decision-making tasks that naturally follow prediction at EGMS scale—such as prioritizing points for inspection, triggering alerts under explicit rule sets, or reasoning under uncertainty and missingness—by leveraging probabilistic logic and neuro-symbolic reasoning frameworks that integrate neural predictors as learned predicates and enable transparent hybrid inference [51]. Total, while symbolic methods are unlikely to replace high-accuracy neural forecasting in this noisy, real-valued setting, a neuro-symbolic layer could materially strengthen the pipeline's trustworthiness, interpretability, and operational

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usefulness by turning learned forecasts into outputs that can be checked, constrained, and acted upon with explicit temporal guarantees [52].

Overall, the findings demonstrate that when the acquisition-driven properties of satellite deformation series are reflected in the learning formulation, short-term deformation forecasting becomes feasible at continental-scale data dimensions. The presented framework therefore provides a reproducible baseline for large-scale deformation prediction, and can serve as a foundation for operational early-evolution monitoring or integration with physically based slope-stability analyses. Future developments may extend the formulation to multi-step horizons or incorporate auxiliary hydrological or lithological indicators; however, the current results provide evidence of the effectiveness and reliability of the proposed incremental-based, temporally consistent learning implementation.

7. Data availability

Code and data are available at the following link.

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